

The Scarlet Witch — Mono-Red Big Mana

Bracket 3 · 3/3 Game Changers · no infinite combos · 100 cards · finalized 2026-08-02

Lands 33 (19 basic)	Instant/Sorceries 39	Discounted (MV 4+) 25	X-spells 5	Creatures 13	Avg MV 3.52	Mana engines 7
Draw 12	Protection 7	Removal/Defence 8	Win conditions 9	Game Changers 3 / 3		

COMMANDER (1)

1 The Scarlet Witch 2 3

LANDS (33)

1 Ancient Tomb 4
1 Valakut, the Molten Pinnacle 2
1 Nykthos, Shrine to Nyx 4
1 War Room 4
1 Rogue's Passage 4
1 Tyrite Sanctum 4
1 Forge of Heroes 4
1 Cori Mountain Monastery 2
1 Arena of Glory 2
1 Shatterskull Smashing 2
1 Valakut Awakening 2
1 Sokenzan, Crucible of Defiance 2
1 Demolition Field 4
1 Scavenger Grounds 4
19 Mountain 2

RAMP — ROCKS AND RITUALS (12)

1 Sol Ring 1 1
1 Arcane Signet 2 2
1 Mind Stone 2 2
1 Ruby Medallion 2 2
1 Tablet of Discovery 2 3
1 Pyromancer's Goggles 5 5
1 Pyretic Ritual 1 2
1 Desperate Ritual 1 2
1 Seething Song 2 3
1 Jeska's Will 2 3
1 Mana Geyser 3 5
1 Brass's Bounty 6 7

RAMP — PER-SPELL MANA ENGINES (7)

1 Birgi, God of Storytelling 2 4 3
1 Electro, Assaulting Battery 1 2 3
1 Urabrask 2 2 4
1 Ashling, Flame Dancer 2 2 4
1 Neheb, the Eternal 3 2 5
1 Storm-Kiln Artist 3 2 4
1 The Vision and Scarlet Witch 2 2 4

CARD DRAW (12)

1 Big Score 3 2 4
1 Unexpected Windfall 2 2 4
1 Pirate's Pillage 3 2 4
1 Inspired Tinkering 4 2 5
1 Commune with Lava X 2 2
1 Ignite the Future 3 2 4
1 Hex Magic 2 2 3
1 Wiccan, Young Avenger 3 2 4
1 Wheel of Fortune 2 2 3
1 Reforge the Soul 3 2 5
1 Will of the Jeskai 3 2 4
1 The One Ring 4 4

THEME / SYNERGY — PUMP HER POWER TO GROW THE DISCOUNT (5)

1 Livaan, Cultist of Tiamat 2 2 3
1 Cait Sith, Fortune Teller 3 2 4
1 Blazing Shoal X 2 2 2
1 Monstrous Rage 2 1
1 Blackblade Reforged 2 2

THEME / SYNERGY — COPY AND RECUR (6)

1 Mizzix's Mastery 3 2 4
1 Past in Flames 3 2 4
1 Increasing Vengeance 2 2 2
1 Repeated Reverberation 2 2 4
1 Arcane Bombardment 4 2 6
1 Improvisation Capstone 5 2 7

PROTECTION (7)

1 Deflecting Swat 2 2 3
1 Bolt Bend 3 2 4
1 Return the Favor 2 2 2
1 Hexing Squelcher 1 2 2
1 Champion's Helm 3 3
1 Mithril Coat 3 3
1 Commander's Plate 1 1

REMOVAL & DEFENCE (8)

1 Chaos Warp 2 2 3
1 Abrade 1 2 2
1 Vandalblast 2 1
1 Untimely Malfunction 1 2 2
1 Zuko's Exile 5 5
1 Fiery Confluence 2 2 4
1 Volcanic Vision 5 2 7
1 Disrupt Decorum 2 2 4

WIN CONDITIONS (9)

1 Crackle with Power X X X 2 2
1 Apex of Power 7 2 2 10
1 Chandra's Ignition 3 2 5
1 Thor, God of Thunder 3 2 5
1 Longshot, Rebel Bowman 3 2 4
1 Fiery Inscription 2 2 3
1 Storm King's Thunder X 2 2 3
1 Electrodominance X 2 2 2
1 Solphim, Mayhem Dominus 2 2 4

Build history — 12 substitutions made to finalize this list

IN 	Ashling, Flame Dancer {2}{R}{R} 4/4. Keeps unspent red mana across steps. Loots on every instant/sorcery; 2nd resolution each turn deals 2 to each opponent and their creatures; 3rd adds {R}{R}{R}{R}.	Fellwar Stone {2} {T}: Add one mana of any colour a land an opponent controls could produce.	 OUT
Why: A build error being corrected. In mono-red, Fellwar Stone often cannot produce {R} at all — and the deck has {R}{R} and {R}{R}{R} costs throughout. Ashling both fixes flood (loot every spell) and produces mana.		Bring back: Tables where every opponent plays red.	
IN 	Champion's Helm {3} Equipped creature gets +2/+2 and, while it's legendary, has hexproof. Equip {1}.	Swiftfoot Boots {2} Equipped creature has hexproof and haste. Equip {1}.	 OUT
Why: A swap, not redundancy — hexproof does not stack. Haste is worth nothing here because Wanda's discount is a STATIC ability that works the moment she resolves. The +2/+2 is +2 off every discounted spell, and 10 of our 12 creatures are legendary.		Bring back: Artifact-removal-heavy pods, or if you want haste for the Blackblade commander-damage line.	
IN 	Disrupt Decorum {2}{R}{R} Goad all creatures you don't control — until your next turn they must attack, and must attack someone other than you.	Blazing Crescendo {1}{R} Target creature gets +3/+1. Exile the top card of your library; you may play it until the end of your next turn.	 OUT
Why: Reversal of an earlier bad cut. The deck had NO way to avoid being attacked, and red gets no fogs. Goad is a one-sided fog that also points the table at each other. Crescendo was outclassed by Monstrous Rage anyway — same +3/+1 for half the mana, plus a permanent +1/+1 Role.		Bring back: If you want a sixth pump effect.	
IN 	Fiery Confluence {2}{R}{R} Choose three, repeats allowed: 1 damage to each creature / 2 damage to each opponent / destroy target artifact.	Insurrection {5}{R}{R}{R} Untap all creatures and gain control of them until end of turn. They gain haste.	 OUT
Why: MV 4, so Wanda drops it to {R}{R} — six damage to each opponent for two mana, or a three-damage sweeper. Insurrection is six mana after the discount, wins through COMBAT (the one axis this deck isn't built for), and is blank against a creature-light table.		Bring back: Creature-heavy boards worth stealing. Note it is NOT a defensive card — it's a sorcery and can't answer an attack.	
IN 	Fiery Inscription {2}{R} Whenever you cast an instant or sorcery, deals 2 damage to each opponent. On entry, the Ring tempts you.	Double Vision {3}{R}{R} Whenever you cast your first instant or sorcery each turn, copy it.	 OUT
Why: Three mana for 2 to each opponent per spell — a five-spell turn is 30 damage across the table — and being an enchantment it survives the removal that kills Thor and Longshot. Double Vision was five mana to copy a single spell per turn.		Bring back: Grindy games where you cast one spell a turn for many turns.	

IN



Hex Magic {2}{R}

Exile your whole hand, draw that many, and you may still play the exiled cards until the end of your next turn.



Rousing Refrain {3}{R}{R}

Add {R} for each card in target opponent's hand, retained until end of turn. Suspend 3—{1}{R}.



OUT

Why: Rousing Refrain scales off your OPPONENTS' hand sizes — dead against a table that has dumped its grip. Hex Magic is reliable, cheaper, and effectively doubles your available cards for the turn.

Bring back: Full-grip tables: control, counterspell decks, Rhystic/Remora pods. Three opponents holding four cards each is 12 red mana.

IN



Mithril Coat {3}

Flash. Indestructible. On entry, attach it to target legendary creature you control. Equipped creature has indestructible.



Blasphemous Act {8}{R}

Costs {1} less for each creature on the battlefield. Deals 13 damage to each creature.



OUT

Why: The deck had no answer to a board wipe: hexproof does nothing (a wrath doesn't target), Commander's Plate's protection doesn't stop a destroy effect and gives nothing vs red, and all three redirects need a spell with a single target. Mithril Coat has FLASH, so you hold it and deploy in response. Blasphemous Act was also the only sweeper that killed Wanda — and the claim it 'pairs with the damage doublers' fails, since 13 already kills everything and it deals no damage to players.

Bring back: Tables with LARGE creatures, where Chandra's Ignition at low power and Fiery Confluence at 3 don't get there.

IN



Neheb, the Eternal {3}{R}{R}

4/6 afflict 3. At the beginning of each of your postcombat main phases, add {R} for each 1 life your opponents have lost this turn.



Comet Storm {X}{R}{R}

Multikicker {1}. Deals X damage to each of up to 1 + kicks targets.



OUT

Why: A Fiery-Confluence turn hands you 18+ red mana in your second main phase. Comet Storm was redundant with Crackle with Power, which does the same job better.

Bring back: When you want a second instant-speed X-spell.

IN



Solphim, Mayhem Dominus {2}{R}{R}

5/4. Doubles all noncombat damage you deal to opponents and to permanents they control. Can give itself an indestructible counter.



Fiery Emancipation {3}{R}{R}{R}

Triples all damage from sources you control.



OUT

Why: Two mana cheaper, and it doubles ONLY damage aimed at opponents — Fiery Emancipation triples damage to your own board too, which would make Blasphemous Act read 39 to each creature. Neither gets Wanda's discount (creature and enchantment), so the two-mana gap is the whole argument. Solphim halves the X you need on Crackle.

Bring back: If you want the triple instead of the double and can afford six mana.

IN



Urabrask {2}{R}{R}

4/4 first strike. Whenever you cast an instant or sorcery, deals 1 to target opponent AND adds {R}. Flips into a Saga once you've cast three in a turn.



Molten-Core Maestro {1}{R}

Opus — whenever you cast an instant/sorcery, put a +1/+1 counter on it. If 5+ mana was spent, add {R} equal to its power.



OUT

Why: Measured, not guessed: Maestro's mana ability needs 5+ mana SPENT, and only 6 of 37 fixed-cost instants/sorceries still cost that much after a 2-power discount. Wanda's discount actively turns it off — the better she is, the less it triggers. Urabrask's {R} is unconditional.

Bring back: Only if the deck somehow stops discounting.

IN



Volcanic Vision {5}{R}{R}

Return an instant/sorcery from your graveyard to hand; deals damage equal to that card's MV to each creature YOUR OPPONENTS control. Exiles itself.



Wild Ricochet {2}{R}{R}

Change targets of target instant/sorcery, then copy it.

OUT



Why: MV 7 → {3}{R}{R}. Return Apex of Power and it's a 10-damage ONE-SIDED wipe plus the regrowth — it doesn't kill your own engines. Wild Ricochet was the fourth redirect behind Deflecting Swat, Bolt Bend and Return the Favor.

Bring back: Spell-heavy pods where redirect-and-copy earns its slot.

IN



Wiccan, Young Avenger {3}{R}

3/3. Whenever you cast a noncreature spell, exile the top card of your library — you may play it until your next end step.



Call Forth the Tempest {5}{R}{R}{R}

Cascade, cascade. Deals damage to each creature opponents control equal to the total MV of your other spells this turn.

OUT



Why: Impulse card advantage on EVERY noncreature spell — a six-spell turn draws six extra cards, for 24 cents. Tempest was eight mana with a random cascade.

Bring back: Creature-heavy boards where the sweep half matters.

Sideboard — what to bring in, and when



Blasphemous Act {8}{R}

Bring in: IN for Volcanic Vision, vs boards with LARGE creatures where Chandra's Ignition at low power and Fiery Confluence at 3 don't get there. Costs {1} less per creature on the battlefield — often {R}. Remember it kills Wanda too.



Blazing Crescendo {1}{R}

Bring in: IN for Monstrous Rage, if you want a sixth pump. Outclassed — Monstrous Rage is the same +3/+1 for half the mana and leaves a permanent +1/+1 Role.



Call Forth the Tempest {5}{R}{R}{R}

Bring in: IN for Apex of Power, vs creature-heavy boards. Cascade twice, then damage to each creature opponents control equal to the total MV of your other spells this turn.



Comet Storm {X}{R}{R}

Bring in: IN for Electrodominance, when you want a second instant-speed X-spell. Multikicker {1} per extra target.



Coruscation Mage {1}{R}

Bring in: IN for Fiery Inscription, vs low-interaction tables where a chip clock survives. Offspring {2} gives a second body. Dies to any wipe.



Darksteel Plate {3}

Bring in: IN for Return the Favor, when you want a SECOND indestructible source. Five mana all-in and no flash, so it's strictly worse than Mithril Coat.



Double Vision {3}{R}{R}

Bring in: IN for Repeated Reverberation, in grindy games where you cast one spell a turn for many turns. Copies only your FIRST instant/sorcery each turn.



Fellwar Stone {2}

Bring in: IN for Mind Stone, only at tables with red opponents. In mono-red it often cannot produce {R} at all, and the deck has {R}{R} and {R}{R}{R} costs throughout.



Fiery Emancipation {3}{R}{R}{R}

Bring in: IN for Solphim, if you want the triple instead of the double. Six mana, an enchantment so NO discount, and it triples damage to your own board too.



Hit the Mother Lode {4}{R}{R}{R}

Bring in: IN for Tablet of Discovery. Discover 10 free-casts your best card. Its Treasures enter TAPPED, so that mana is for next turn — and Brass's Bounty makes ~14 untapped for the same 5 mana.



Insurrection {5}{R}{R}{R}

Bring in: IN for Chandra's Ignition, vs creature-heavy boards worth stealing. NOT a defensive card — it's a sorcery, so it cannot answer an attack.



Molten-Core Maestro {1}{R}

Bring in: IN for Storm-Kiln Artist, only if the deck somehow stops discounting. Its mana ability needs 5+ mana SPENT, and only 6 of 37 fixed-cost spells cost that after a 2-power discount — Wanda actively turns it off.



Pinnacle Monk {3}{R}{R}

Bring in: IN for a Mountain, if you keep flooding or getting land-screwed. The spell half is a CREATURE, so Wanda never discounts it, and Mystic Peak isn't a Mountain — it turns off Valakut and adds no Nykthos devotion.



Rousing Refrain {3}{R}{R}

Bring in: IN for Mana Geyser, vs full-grip tables — control, counterspell decks, Rhystic Study / Mystic Remora pods. Three opponents holding four cards each is 12 red mana. Dead against aggro.



Wild Ricochet {2}{R}{R}

Bring in: IN for Untimely Malfunction, vs spell-heavy pods. Redirect AND copy. It was the fourth redirect behind Deflecting Swat, Bolt Bend and Return the Favor.



Jaya's Immolating Inferno {X}{R}{R}

Bring in: IN for Electrodominance, when you want a second Crackle. X damage to up to three targets, all different. The legendary requirement is always met.



Nova Flame {X}{2}{R}{R}

Bring in: IN for Volcanic Vision, vs creature decks. The +1/+1 counters on Wanda are PERMANENT, so the discount grows — but it deals no damage to players and can't close.



Prisoner's Dilemma {3}{R}{R}

Bring in: IN for Fiery Confluence, vs pods that enjoy politics. 4/8/12 damage to each opponent, but THEY choose the outcome.



Swiftfoot Boots {2}

Bring in: IN for Champion's Helm, vs artifact-removal-heavy pods, or when you want haste for the Blackblade commander-damage line. Hexproof does not stack.



Witch's Mark {1}{R}

Bring in: IN for Monstrous Rage, in grindy tables where the extra card beats tempo. MV 2 so no discount, but the Wicked Role is a permanent +1 to Wanda's power.

Bracket 4 upgrade path

Swap these 14 and the Bracket 3 list becomes the Bracket 4 list in DECK-B4.md — 86 cards stay the same. The fair mana base becomes a fast one and Underworld Breach turns the graveyard into a second hand. Everything costing 5+ that isn't a payoff comes out. Game Changers go from 3 to 10, which is legal at Bracket 4 and NOT at Bracket 3.

B4



Chrome Mox {0}

Imprint: exile a nonland, nonartifact card from hand. {T}: Add one mana of the imprinted card's colour.

Pyromancer's Goggles {5}

{T}: Add {R}. When that mana is spent on a red instant/sorcery, copy that spell.

B3



Why: Goggles is a five-mana investment that pays off a turn later. B4 needs turn-one acceleration more than a delayed copy engine.

Revert to B3 when: Always — Game Changer. Goggles is one of the best cards in the fair build: the copy keeps X and can re-use targets.

B4



City of Traitors —

{T}: Add {C}{C}. When you play another land, sacrifice this.

Mountain —

Basic Land — Mountain.

B3



Why: Two mana off one land pulls the Breach turn forward by a full turn cycle.

Revert to B3 when: The fair build wants stable mana more than a burst, and Valakut counts Mountains.

B4



Dance with Calamity {7}{R}

Exile cards from the top one at a time; while the total MV stays 13 or less you may cast any number of them free.

Fiery Inscription {2}{R}

Whenever you cast an instant or sorcery, deals 2 damage to each opponent.

B3



Why: A chip clock needs several turns and B4 doesn't have several turns. Dance converts a big mana pile straight into a board.

Revert to B3 when: Fiery Inscription is 30+ damage over a normal fair-length game.

B4



Faithless Looting {R}

Draw two cards, then discard two. Flashback {2}{R}.

Solphim, Mayhem Dominus {2}{R}{R}

Doubles noncombat damage you deal to opponents and their permanents.

B3



Why: Looting fills the graveyard for Underworld Breach, so discarding is upside. Solphim does nothing on an empty board.

Revert to B3 when: Solphim halves the X you need on Crackle, which is the fair build's entire kill.

B4



Finale of Promise {X}{R}{R}

Cast up to one instant and one sorcery from your graveyard with MV X or less, free. If X is 10+, copy each twice.

Disrupt Decorum {2}{R}{R}

Goad all creatures you don't control.

B3



Why: B4 wins before the crack-back matters, so the fog is worth less than a second recursion spell.

Revert to B3 when: Disrupt Decorum is the fair build's ONLY way to not be attacked.

B4

**Gamble {R}**

Search your library for a card, put it into your hand, then discard a card at random.

**Improvisation Capstone {5}{R}{R}**

Exile from the top until total MV 4+, cast any number free. Paradigm: a free copy from exile at the start of each of your first main phases.

B3



Why: A one-mana tutor for Breach or Crackle. The random discard barely matters once Breach can escape whatever you lose.

Revert to B3 when: Always — Game Changer. Capstone's Paradigm is a recurring engine the fair build wants.

B4

**Grim Monolith {2}**

{T}: Add {C}{C}{C}. Doesn't untap during your untap step; {4}: Untap.

**Tablet of Discovery {2}{R}**

ETB mill a card, may play it this turn. {T}: Add {R}. {T}: Add {R}{R}, instants and sorceries only.

B3



Why: Net +1 mana the turn it lands instead of a three-mana rock that produces two.

Revert to B3 when: Always — Game Changer.

B4

**Lion's Eye Diamond {0}**

{T}, Discard your hand, Sacrifice: Add three mana of any one colour.

**Blackblade Reforged {2}**

Equipped creature gets +1/+1 for each land you control. Equip legendary {3}, equip {7}.

B3



Why: LED is the other half of Underworld Breach — discarding your hand is upside once Breach is out. Blackblade is a five-mana all-in that deals no damage the turn it lands.

Revert to B3 when: Always — Game Changer. Blackblade is the biggest single discount boost in the fair build.

B4

**Lotus Petal {0}**

{T}, Sacrifice: Add one mana of any colour.

**Monstrous Rage {R}**

Target creature gets +2/+0 and a Monster Role token (+1/+1, trample).

B3



Why: The pump package shrinks to three in B4 — the fast build wins before growing Wanda pays off.

Revert to B3 when: The permanent +1/+1 Role is a permanent discount in the fair build.

B4

**Mana Vault {1}**

{T}: Add {C}{C}{C}. Doesn't untap normally; {4}: Untap. 1 damage per upkeep it stays tapped.

**Mind Stone {2}**

{T}: Add {C}. {1},{T}, Sacrifice: Draw a card.

B3



Why: Bracket 4 wants explosive mana on turns 1–3, not a fair rock that ramps by one. Three mana off a one-drop.

Revert to B3 when: Always — Mana Vault is a Game Changer and B3 caps at three.

B4

**Mox Amber {0}**

{T}: Add one mana of any colour among legendary creatures and planeswalkers you control.

**Reforge the Soul {3}{R}{R}**

Each player discards their hand, then draws seven. Miracle {1}{R}.

B3



Why: Wanda switches Mox Amber on by herself, and B4 already runs Wheel of Fortune, Will of the Jeskai and Faithless Looting.

Revert to B3 when: The miracle cost makes Reforge a two-mana Wheel whenever you draw it first — watch every draw step.

B4



Mox Diamond {0}

If this would enter, discard a land card instead or bin it. {T}: Add one mana of any colour.



Brass's Bounty {6}{R}

Create a Treasure for each land you control — untapped.

B3



Why: Brass's Bounty is 5 mana after the discount for ~14 untapped Treasures, which is enormous but a turn-7 card. B4 is trying to win by then.

Revert to B3 when: Always — Game Changer. Brass's Bounty is the fair build's biggest single mana burst.

B4



Reiterate {1}{R}{R}

Buyback {3}. Copy target instant or sorcery spell.



Zuko's Exile {5}

Exile target artifact, creature or enchantment. Its controller creates a Clue.

B3



Why: Reiterate with buyback plus Mana Geyser is infinite mana, and it copies X-spells at full size.

Revert to B3 when: ALWAYS. This is the single card that would make the Bracket 3 list an infinite combo deck — it must never appear there.

B4



Underworld Breach {1}{R}

Each nonland card in your graveyard has escape — its mana cost plus exile three other cards.

Sacrifice it at end step.



Arcane Bombardment {4}{R}{R}

First instant/sorcery each turn: exile one at random from your yard, then copy every card exiled with it, free.

B3



Why: Breach turns the graveyard into a second hand, and because escape cost equals the mana cost, X-SPILLS ESCAPE AT FULL SIZE. Arcane Bombardment's copies are free-cast, so they force X=0.

Revert to B3 when: Always — Game Changer, and it enables lines a fair pod won't enjoy.

Discount = Wanda's power + Ruby Medallion + Longshot, generic only, MV 4+ only. Crackle: X costs 3 generic each and deals 5X. X=8 is 40 = a kill from full. Free-casting an X-spell forces X=0. Full pilot notes in research/gameplan.md and research/formulas.md.

SCARLET WITCH — PILOT'S GAMEPLAN

How to actually play the deck, turn by turn and board state by board state. Written for the locked 100 in `DECK.md`. Companion math sheet: `formulas.md`.

0. The one-paragraph version

You are **not** a creature deck and you are **not** a storm deck. You are a **cost-reduction engine that fires one enormous spell**. Wanda makes every instant and sorcery of mana value 4+ cheaper by her power, so your whole game is: survive, grow her power, generate a pile of red mana in a single turn, then convert that mana into damage with an X-spell. Every card in the deck is there to do one of four jobs — **make mana, draw cards, grow Wanda, or be the payoff**. If a play doesn't advance one of those four, don't make it.

The mental reframe: stop asking "what can I cast?" Ask "**how much does everything cost right now, and how much mana can I make this turn?**" The gap between those two numbers is your whole deck.

1. The 60-second cheat sheet

START OF TURN

- say the DISCOUNT out loud: "Wanda X + Ruby 1 + Longshot 1 = N off generic"
- did I draw Reforge the Soul first? MIRACLE it for {1}{R}
- The One Ring: tap to draw (remember the burden counters)

PRECOMBAT MAIN

1. Wanda first if she isn't out — nothing matters more
2. Per-spell mana engines (Electro / Birgi / Urabrask / Storm-Kiln)
3. Livaan / pump — TARGET WANDA
4. Card draw
5. Rituals last
6. If Neheb is out: deal damage NOW (Fiery Confluence, pingers)

COMBAT

- usually don't attack; just pass through to reach postcombat

POSTCOMBAT MAIN

- Neheb trigger: {R} per 1 life opponents lost this turn (EMPTIES at end of phase)
- THE X-SPELL GOES HERE
- $(\text{mana} - 2 \text{ red pips} + \text{total reduction}) \div 3 = X$
- 5X is the damage. 40 = a kill from full. Solphim halves the X you need.

HOLD FOREVER

- Deflecting Swat (free with commander)
- Mithril Coat (flash — for the wrath)
- Blazing Shoal (until you have an uncastable fat red card)
- Mana Geyser (until they're tapped out)

NEVER

- free-cast an X-spell ($X = 0$)
- cast Crackle "big but not lethal"
- wheel a good hand

2. The one formula that governs everything

Every instant or sorcery you cast at mana value 4 or greater costs:

printed cost — Wanda's power — 1 (Ruby Medallion) — 1 (Longshot) ... generic only

Reductions **only eat generic mana**. The coloured pips (🔴🔴 on Crackle, 🔴🔴🔴 on Apex) are untouchable. Everything under mana value 4 gets **no discount from Wanda at all** — Ruby Medallion and Longshot still apply.

Say the number out loud before every big spell. In your playtest you forgot Wanda's own 3 and cast Crackle at X=7 instead of X=9. That's 10 damage per target, thrown away.

3. What actually kills people, ranked

A. The X-spell (your primary win con). One spell, whole table.

Card	What it does	The number that matters
Crackle with Power	5×X damage to each of up to X targets	X=8 is 40 — a kill from full
Jaya's / Comet Storm (<i>sideboard</i>)	X to three / X+kicks targets	backup Crackles
Chandra's Ignition	Wanda's power to each other creature and each opponent	needs Wanda pumped; also a one-sided-ish wipe
Apex of Power	10 mana of any one colour + cast 7 free cards	it's a <i>mana engine</i> that also wins

B. Per-spell chip damage. These turn a long turn into a kill without a payoff card:

Card	Trigger	Damage
Thor, God of Thunder	each noncreature spell you cast	equal to that spell's mana value — 4 to 11 a pop
Longshot, Rebel Bowman	each noncreature spell you cast	2 to each opponent, <i>and</i> 1 off everything
Fiery Inscription	each instant/sorcery you cast	2 to each opponent

C. The multiplier. **Solphim** doubles all noncombat damage you aim at opponents and their permanents. It halves the X you need. Nothing else in the deck is a multiplier — one is enough.

D. Commander damage (backup only). Blackblade Reforged plus Rogue's Passage. Take this line only when the X-spell plan is dead.

4. The three numbers you track

Everything else is noise. Track these:

Number 1 — WANDA'S POWER (the discount)

Base 2. It goes up from:

- **Livaan** — +X/+0 where X is the mana value of the noncreature spell you just cast. ****Always**

point it at Wanda. **** Cast a 5-drop, she's +5 for the turn.**

- **Cait Sith** — +X/+0 each combat, X = the exiled card's mana value.
- **Blazing Shoal** — free: exile a red card of mana value X from hand, +X/+0. Ceiling is +10.
- **Monstrous Rage** — +3/+1 and a **permanent** +1/+1 Role.
- **Blackblade Reforged** — +1/+1 per land. Equip is 3 because she's legendary, not 7.
- **Forge of Heroes / Tyrite Sanctum** — permanent +1/+1 counters.

Permanent pumps compound. A Forge counter is a mana saved on every X-spell for the rest of the game. One-turn pumps only matter on the turn you go off.

Number 2 — RED MANA AVAILABLE THIS TURN

Lands + rocks + rituals + per-spell engines. The per-spell engines are the ones people undercount:

- **Electro, Assaulting Battery** — ⚡ per instant/sorcery, ****and you keep all unspent red mana**

across steps and phases**

- **Birgi** — ⚡ per *spell* (creatures and artifacts too), kept until end of turn
- **Urabrask** — ⚡ per instant/sorcery, plus 1 damage
- **The Vision and Scarlet Witch** — ⚡ per spell, and it grows itself
- **Storm-Kiln Artist** — a Treasure per instant/sorcery cast **or copied**
- **Neheb** — ⚡ per 1 life your opponents lost this turn, at your **postcombat** main

Number 3 — CARDS LEFT IN HAND

You run out of cards long before you run out of mana. That's what **Wiccan**, **The One Ring**, **Hex Magic**, **Big Score**, **Commune with Lava** and the wheels are for. If you're at 15 mana and 1 card, you have a card problem, not a mana problem.

5. Turn-by-turn: the opening

T1–T3 — do nothing flashy

Land, Sol Ring, Arcane Signet, Mind Stone, Ruby Medallion. **Cast Wanda around turn 3.** She's a 2/3 that does nothing visible, which is exactly what you want — nobody removes her early because she looks harmless. Every turn she's out is a turn your spells are 2 cheaper.

Do not cast a ritual on turn 2 just because you can. Rituals are for the turn you go off.

T4–T5 — build the discount, find cards

Priorities in order: a **per-spell mana engine** (Electro, Birgi, Urabrask) > **Wiccan / The One Ring** for cards > a **pump** on Wanda > a payoff you're holding.

This is also when you cast **Fiery Inscription** or **Longshot** — they're cheap, they accumulate, and they make the eventual turn lethal instead of merely large.

T6+ — look for the turn

You want, all at once: **12+ mana available, 3+ cards in hand, a payoff, and one protection spell up.** See §8.

Mulligans

Keep any hand with 3+ lands and either a rock or Wanda-plus-a-cheap-engine. **Ship** hands with 5+ lands and no engine, or 2 lands and no rock — this deck's curve genuinely needs 4–5 lands and you can't operate on 2. **Never keep** a hand whose only action is a 7-drop. You will die holding it.

6. The HOLD list — "don't cast this yet"

This is the section that separates a good pilot from a bad one. Every card below is a trap if cast on curve.

Card	HOLD unless...	Why
Blazing Shoal	you have a red card of mana value 6+ in hand that you can't cast this turn	It's free, but it costs two cards. Pitching Apex of Power for +10 is great; pitching a 2-drop for +2 is a waste. Never pitch a card you could actually cast this turn.
Mana Geyser	your opponents are tapped out	It counts <i>their</i> tapped lands. Cast it in your main phase after a big turn cycle, not on an empty board. It's often 0–3 mana on turn 4 and 12+ on turn 8.
Rousing Refrain <i>(sideboard)</i>	opponents are holding cards	Scales off <i>their</i> hand size. Dead against a table that's dumped its grip.
Wheel of Fortune / Reforge the Soul	your hand is 2 cards or fewer	You're refilling <i>yourself</i> . Wheeling with 5 good cards hands three opponents a fresh 7 and gains you nothing.
Reforge the Soul	<i>(exception)</i> you draw it as your first card of the turn	Miracle 1 — cast it immediately for two mana. Watch for this every draw step.
Deflecting Swat	someone targets you or Wanda	It's free with your commander out. There is no reason to ever cast it proactively. Hold it every single turn.
Mithril Coat	a wrath or a removal spell is on the stack	It has flash . Casting it on your own turn wastes its entire point.
Bolt Bend	Wanda has 4+ power	It costs 3 less with a 4-power creature — so 2 instead of 3. Pump first, then it's nearly free.
Chandra's Ignition	Wanda's power is 6 or more	At base 2 power it deals 2. It's a finisher, not a removal spell. Pump, <i>then</i> Ignite.
Blackblade Reforged	you have 5 spare mana and a plan	2 to cast plus 3 to equip is a whole turn that deals no damage. Deploy it on a turn you weren't going to go off anyway.
Disrupt Decorum	you're about to be attacked, or you're the archenemy	Goad lasts until your next turn , so cast it in <i>your</i> main phase to cover the whole cycle back to you.
Past in Flames	your graveyard has 3+ spells worth recasting	It's a payoff, not a cantrip. Casting it with an empty yard is a wasted card.
Improvisation Capstone	you're going to have main phases left to use the Paradigm copies	The free copy each first main phase is most of the card's value.
The One Ring	you need the protection turn, or you need cards	" <i>Protection from everything until your next turn</i> " is a fog for the whole table's attacks . Sometimes the right play is casting it purely to not die.
Vandalblast	you can overload it (4) profitably	Single-target for 2 is fine in a pinch, but overloading is a blowout. Don't fire it at a Sol Ring.
Ancient Tomb	you actually need the second mana	It deals you 2 every activation. At 40 life that's fine; at 12 it's a real cost. Tap Mountains first when the extra mana doesn't change what you cast.

And the mirror image — the CAST IT NOW list

- **Wanda herself.** Every turn she isn't out is a turn everything costs 2 more.
- **Ruby Medallion, Longshot, Fiery Inscription.** Cheap, permanent, compounding.
- **Livaan.** It's 3 mana and it turns every subsequent spell into a discount.
- **Reforge the Soul off the top** — miracle for 1.

7. The X-spell rules you must not forget

Three categories, and they behave completely differently:

✔ **Paying a real cost** → **X works.** Hard-casting. Flashback and escape where the cost equals the mana cost — **Past in Flames, Will of the Jeskai, Underworld Breach** (B4). Wanda still discounts these.

✓ **Copying a spell on the stack** → **X carries over, free. Pyromancer's Goggles, Increasing Vengeance, Repeated Reverberation, Return the Favor, Storm King's Thunder.** The copy keeps your X and can **re-use the same targets** — the "no duplicate targets" rule is per spell, and a copy is a separate spell.

✗ **Free-casting** → **X = 0, deals nothing. Apex of Power, Improvisation Capstone, Mizzix's Mastery, Electrodominance, Arcane Bombardment.** If Apex exiles Crackle with Power, **do not cast it** — put it in hand if the effect allows, or cast something else.

The one-liner: if you're paying a cost, X works. If you're paying nothing, X is zero.

8. The go-off checklist — do I start this turn?

Before you cast the first ritual, confirm **all five**. If any is missing, don't start — you'll dump your hand and die holding an empty board.

1. **Mana.** Count lands + rocks + rituals + engine output. Write the number down.
2. **Cards.** 3+ in hand, or a draw spell you can cast first (Wiccan out counts double).
3. **The payoff.** Crackle / Ignition / Thor in play / a copy effect on a payoff. Naming which one.
4. **The number.** Do the division: $(\text{mana} - 2 \text{ red pips}) \div 3 = \text{your X}$. Is 5X enough?
5. **The window.** One protection spell up — Deflecting Swat (free), Mithril Coat, or Bolt Bend —

or the knowledge that nobody at the table is holding interaction.

If you can kill exactly one player, that's often still correct — but kill the one who's closest to winning, not the one who's most annoying.

9. Sequencing — order of operations within your turn

The order matters more in this deck than almost any other. The default:

1. PRECOMBAT MAIN
 - a. Cast Wanda if she isn't out. Nothing else matters more.
 - b. Deploy per-spell mana engines FIRST (Electro / Birgi / Urabrask / Storm-Kiln). Every spell after this point makes mana.
 - c. Deploy pump (Livaan) BEFORE the expensive spells, so they grow her on the way.
 - d. Cheap card draw (Big Score, Wiccan already out) to see more of the deck.
 - e. Rituals LAST — they're mana you can't hold, so spend them into the payoff.
 - f. If Neheb is out: deal your damage HERE (Fiery Confluence, pinger triggers).
2. COMBAT

Usually you don't attack. Go to combat anyway — the phase happens regardless and you need to reach your postcombat main.
3. POSTCOMBAT MAIN
 - g. Neheb triggers: {R} for each 1 life your opponents lost this turn.
 - h. THIS is where the X-spell goes. Count your reducers, do the division, cast it.

Why Neheb rewrites the turn

Neheb gives red mana at the **beginning of your postcombat main phase**, equal to total life your opponents lost that turn. So the shape is: **burn precombat, collect postcombat, kill postcombat.**

That mana **empties at end of phase** — you cannot hold it into your end step or anyone's turn. Use it or lose it.

Two Goggles reminders

- Tap **Pyromancer's Goggles** and spend *that specific* 🗑 on the spell, or no copy.
- Goggles produces one 🗑 per turn, so **one copy per turn cycle** — unlike the copy *spells*,

which stack freely.

10. Scenario library

🎯 Scenario 1 — Turn 6, 14 mana available, Crackle in hand, Wanda at 2

Reducers: Wanda 2, Ruby Medallion 1. That's 3 off generic.

$14 \text{ mana} - 2 \text{ (red pips)} = 12 \text{ generic. } 12 + 3 \text{ reduction} = 15 \text{ generic budget. } 15 \div 3 = X=5.$

X=5 is **25 damage to each of up to 5 targets** — 75 across the table. Not a kill from 40, but it puts everyone in one-shot range and you've spent one card.

Better line: don't cast it. Spend the turn on Livaan + a mana engine + a draw spell, and cast Crackle next turn at X=8 for the actual kill. **Crackle is at its worst as a "big but not lethal" spell** — it announces you as the threat and gets you killed before your next turn.

🌟 Scenario 2 — Your playtest board (the real one)

Wanda at 3, Ruby Medallion, Longshot, Goggles, Neheb, 24 mana in your postcombat main.

Reducers: $3 + 1 + 1 = 5$ off.

$24 - 2 \text{ red pips} = 22 \text{ generic}$. $22 + 5 = 27 \text{ budget}$. $27 \div 3 = X=9$.

X=9 is 45 damage to each of up to 9 targets. You cast X=7 for 35. The habit that fixes it: say "Longshot one, Ruby one, Wanda three — five off" out loud before you lock X.

🔧 Scenario 3 — You have Goggles and a modest Crackle

Goggles copies the spell and the copy **keeps X and can re-use targets**. So X=2 with Goggles is $10 + 10$ to the *same* two players — **20 each** — for about 4 mana after reducers.

This is your best *early* Crackle. Two players at 20 is a completely different game than three players at 40, and it costs almost nothing.

With Solphim out, that same line is 40 to each of two players — two people dead for four mana.

🛡️ Scenario 4 — A wrath is on the stack

Hexproof (Champion's Helm) and protection (Commander's Plate) **both do nothing** — a wrath neither targets nor deals damage.

Your answers, in order: flash in **Mithril Coat** (this is the entire reason it's in the deck) → **Tyrite Sanctum**'s indestructible counter if you set it up → accept it and rebuild.

If Wanda dies, she's cheap to recast (the tax climbs by 2 each time) and **you should recast her immediately** — the discount is worth more than any other 3-drop.

🔪 Scenario 5 — You're about to be attacked by a big board

Disrupt Decorum. Goad everything you don't control; they must attack, and must attack someone other than you. It buys the exact turn you need. Cast it in *your* main phase — goad runs until your next turn, so it covers the whole cycle.

Second option: **The One Ring** for protection from everything until your next turn. That's a fog for the entire table and it draws you cards afterwards.

Third: **Volcanic Vision** returning a big spell — damage equal to its mana value to each creature *your opponents control*, one-sided, and it doesn't kill your engines.

🗨️ Scenario 6 — Wanda has been killed twice and the tax is 6

Don't chase her. At $2\text{🔴} + 4$ tax she's a 7-mana 2/3.

Play the deck without the discount for a turn. Your cheap spells still work — Fiery Confluence is 2🔴🔴 , Big Score is 3🔴 , Longshot and Thor still ping. Recast her when it's convenient, or when you're going off anyway and the 3 discount pays for the tax within the same turn.

🌪️ Scenario 7 — You resolve Apex of Power and see Crackle with Power in the exile

Do not cast it. Free-casting forces X=0. Cast the other six cards, and if Apex's wording gives you the choice, take Crackle to hand instead. The 10 mana Apex adds is what makes *next* turn's hard-cast Crackle enormous.

🗨️ Scenario 8 — Big graveyard, Will of the Jeskai in hand

Will of the Jeskai gives everything in your yard **flashback equal to its mana cost** — which for an X-spell **includes X**, so you get to choose a real X.

Best line: cast the X-spell once, let it hit the yard, *then* Will of the Jeskai and flash it back bigger with the mana you've accumulated. You get two X-spells out of one card.

Also: with your commander out, Will of the Jeskai takes **both** modes. Don't forget the wheel half.

🏠 Scenario 9 — You're flooded: 20 mana, 1 card in hand

You have a **card** problem. Do not cast the ritual.

Priority: **The One Ring** (draws increasingly many) → **Hex Magic** (doubles your hand for a turn) → **Commune with Lava** for a big X → **War Room** → **Apex of Power** (7 free cards *and* 10 mana).

Apex is the best flood-breaker in the deck precisely because it converts excess mana into cards.

🔥 Scenario 10 — Someone else is about to win

Your interaction is thin and mostly proactive. In rough order:

Chaos Warp the problem permanent (only unconditional answer) → **Untimely Malfunction** to redirect their kill spell or blank an attack → **Deflecting Swat** (free!) to redirect the spell that wins the game → **Vandalblast** overloaded if it's an artifact deck → **Disrupt Decorum** to turn their board on each other.

Do not hold your X-spell hoping to race. If you can't kill the table, kill *them*.

🎯 Scenario 11 — Choosing targets for Crackle

Targets must all be **different** — you cannot double up on one player. And **damage per target is fixed at 5X regardless of how many you choose**, so picking fewer doesn't concentrate it.

Order of preference: **players closest to winning** → **players in range of your chip damage next turn** → **problem creatures** you can't otherwise remove. Remember that "any target" includes planeswalkers and battles.

Scenario 12 — Turn 4, you drew Reforge the Soul as your first card

Miracle it for 1  **immediately.** Everyone discards and draws seven. You're the one with a mana engine and a 3-mana commander; a fresh seven cards is worth far more to you than to them.

Scenario 13 — Thor and Longshot are both out

Every noncreature spell you cast now deals **its mana value (Thor) plus 2 to each opponent (Longshot)**. A five-spell turn averaging mana value 5 is 25 damage focused plus 30 spread — before any payoff.

This is when you don't need Crackle at all. Chain cheap spells and let the triggers kill. Also note: **Goggles copies do NOT trigger these** — a copy isn't cast.

Scenario 14 — Solphim is on the battlefield

Halve every damage number you were planning. X=4 Crackle becomes 40 to each of four. Fiery Confluence becomes 12 to each opponent for two mana. Chandra's Ignition on a 6-power Wanda becomes 12 to each opponent *and* 12 to every other creature.

With Solphim out, look for the kill a full turn earlier than you otherwise would.

Scenario 15 — You have Valakut, the Molten Pinnacle and 5+ other Mountains

Every Mountain you play from here deals **3 damage to any target**. That's a free 3 a turn, it feeds Neheb, and it closes games you thought were out of reach. Count it before deciding you can't get there — and if you're holding two Mountains, that's 6.

11. Threat assessment — who do you point at?

You get **one** big turn. Spend it on the right person.

1. **Whoever is closest to winning**, even if they're at high life. You're the only one who can hit

all three at once; use that on the person a normal deck can't answer.

1. **Whoever is holding up blue mana**, if you're forced to choose. Your combo turn dies to one

counterspell and you only have Hexing Squelcher for that.

1. **Whoever has been attacking you.** Practical, not emotional — dead attackers stop the clock.
2. **Never** spend a whole turn killing the weakest player. You're not going to get a second one.

12. The checklist people forget

Before every big spell:

- ☐ **Wanda's power** — did you count it? (This is the one you missed.)
- ☐ **Ruby Medallion 1** · **Longshot 1**
- ☐ Is the spell **mana value 4+**? If not, Wanda does nothing.
- ☐ Is it an **X-spell being free-cast**? If so, X = 0 — don't.
- ☐ Did you **tap Goggles and spend that 2 on this spell**?
- ☐ **Livaan** — target Wanda, not something else.
- ☐ **Thor / Longshot / Fiery Inscription** — did you resolve their triggers on *every* spell?
- ☐ **Storm-Kiln Artist** makes a Treasure on cast **and on copy**.
- ☐ **Ashling** — second resolution each turn is 2 damage to each opponent, third is .
- ☐ Neheb mana **empties at end of phase**.

13. Beginner mistakes to avoid

1. **Forgetting Wanda's own discount.** Costs you an entire X. Say the reducers out loud.
2. **Free-casting an X-spell.** Deals zero. The single worst blowout available to you.
3. **Casting Crackle "big but not lethal."** You become the archenemy and die before your next

turn. Either kill someone or wait.

1. **Casting rituals on curve.** Rituals are turn-of-the-kill cards, not ramp.
2. **Wheeling a good hand.** You're refilling three opponents for free.
3. **Casting Deflecting Swat proactively.** It's free — hold it forever.
4. **Casting Mithril Coat on your own turn.** It has flash. Hold it.
5. **Chandra's Ignition at 2 power.** Deals 2. Pump first.

6. **Attacking with Wanda for no reason.** She's a 2/3 that runs your entire deck. Leave her home

unless Blackblade + Rogue's Passage is the actual plan.

1. **Pitching a castable card to Blazing Shoal.** Only pitch what you can't cast.
2. **Not counting Valakut triggers.** Three damage a Mountain adds up.
3. **Tapping Ancient Tomb when Mountains would do.** Two life a pop is real.

SCARLET WITCH — COST & DAMAGE FORMULAS

Companion to `gameplan.md`. Every card in the locked 100 that produces a number, written as a formula you can evaluate at the table.

1. Symbols

Symbol	Meaning
W	The Scarlet Witch's power (base 2)
R	Total generic cost reduction = W + 1 (Ruby Medallion) + 1 (Longshot)
MV	Mana value of the spell as it sits on the stack (X counts at its announced value)
X	The value you announce for an X-spell
S	Spells you've cast this turn
L	Lands you control
P	Opponents still alive — normally 3
Λ	Total life your opponents have lost this turn (all of them combined)
D	Your devotion to red — red pips on permanents you control

×2 if Solphim is out applies to every noncombat damage number aimed at an opponent or a permanent an opponent controls. It does **not** double damage to you or your own creatures.

2. The master cost formula

Cost of an instant/sorcery = printed cost − R	(generic portion only)
R = W + 1 (Ruby Medallion) + 1 (Longshot)	

Wanda's share only applies at MV ≥ 4. Ruby Medallion (red spells) and Longshot (noncreature spells) apply at every mana value. Coloured pips can never be reduced.

Board state	R (generic off)
Wanda alone, base power	2
Wanda + Ruby Medallion	3
Wanda + Ruby + Longshot	4
Wanda at 3 (one Forge counter) + Ruby + Longshot	5
Wanda at 6 (Livaan off a 4-drop) + Ruby + Longshot	8

3. Crackle with Power — the master table

xxx🔥 → **5×X damage to each of up to X *different* targets.**

Mana needed = 3X + 2 - R	
Max X you can afford = [(available mana - 2 + R) ÷ 3]	
Damage per target = 5X	(×2 with Solphim)
MV on the stack = 3X + 2	

X	Raw cost	R=3	R=4	R=5	R=8	Damage each	Targets
2	8	5	4	3	—	10	2
3	11	8	7	6	3	15	3
4	14	11	10	9	6	20	4
5	17	14	13	12	9	25	5
6	20	17	16	15	12	30	6
7	23	20	19	18	15	35	7
8	26	23	22	21	18	40 ← kill from full	8
9	29	26	25	24	21	45	9
10	32	29	28	27	24	50	10

The two numbers to memorise: every point of X costs **3 more generic**, and **X=8 is the kill** (40 damage). With **Solphim**, X=4 is the kill instead — that's 9 mana at R=5.

Targeting rules

- Targets must all be **different** (rule 601.2c). No doubling up on one player.
- Damage per target is **fixed at 5X** regardless of how many targets you pick. Choosing fewer

does *not* concentrate it.

- To put 20 on one player you need **X=4** and simply choose one target.

4. The three X rules

Situation	X	Cards
Paying a real cost — hard-cast, or flashback/escape where the cost = the mana cost	✔ you choose	Past in Flames, Will of the Jeskai, Underworld Breach (B4)
Copying a spell on the stack — the copy inherits X and can re-use targets	✔ free, same X	Pyromancer's Goggles, Increasing Vengeance, Repeated Reverberation, Return the Favor, Storm King's Thunder
Cast "without paying its mana cost"	✗ X = 0	Apex of Power, Improvisation Capstone, Mizzix's Mastery, Electrodominance, Arcane Bombardment, Dance with Calamity (B4), Finale of Promise (B4)

Rule 107.3b: X is forced to 0 only when you pay *neither the mana cost nor an alternative cost that includes X*. Flashback and escape costs that equal the mana cost **include** X, so they're fine. Rule 707.10: a copy inherits "modes, targets, **the value of X**, and additional or alternative costs."

5. Mana engines — output per turn

Card	Output	Notes
Electro, Assaulting Battery	+1 ⚡ per instant/sorcery	Keeps ALL unspent red mana across steps and phases — the best of the group
Birgi	+1 ⚡ per spell	Any spell, creatures included. Kept until end of turn.
Urabrask	+1 ⚡ per instant/sorcery	Plus 1 damage to an opponent
The Vision and Scarlet Witch	+1 ⚡ per spell	Also grows itself — a real clock
Storm-Kiln Artist	+1 Treasure per instant/sorcery cast or copied	The only engine that counts <i>copies</i>
Ashling, Flame Dancer	loot per spell; 2nd resolution = 2 dmg to each opponent + their creatures; 3rd = ⚡⚡⚡⚡	Also keeps unspent red mana
Neheb, the Eternal	+Λ ⚡ at your postcombat main	Empties at end of that phase
Nykthos	2, ⚡: add D red	Devotion to red; a real board is D = 6–10
Mana Geyser	+1 ⚡ per tapped land opponents control	0 on an empty board, 12+ after a turn cycle
Jeska's Will	with commander: both — ⚡ per card in an opponent's hand and exile 3, play them this turn	
Brass's Bounty	+L Treasures, untapped	~14 at 5 mana after reduction
Rousing Refrain (SB)	⚡ per card in target opponent's hand, kept	
Rituals	Pyretic/Desperate ⚡⚡⚡ (net +1) · Seething Song ⚡⚡⚡⚡⚡ (net +2)	Cast on the kill turn only

Total per-spell yield with Electro + Birgi + Urabrask + The Vision all out: **4 red mana per instant/sorcery**, plus a Treasure from Storm-Kiln. At that point cheap spells are free and expensive ones are nearly free.

6. Damage sources as formulas

Card	Formula	Per opponent?
Crackle with Power	5X to each of up to X targets	choose
Chandra's Ignition	W to each other creature and each opponent	yes — and Wanda survives it
Thor, God of Thunder	MV of each noncreature spell you cast , to any target	one target, focusable
Longshot, Rebel Bowman	2 per noncreature spell cast	yes, each opponent
Fiery Inscription	2 per instant/sorcery	yes, each opponent
Urabrask	1 per instant/sorcery	one opponent
Ashling (2nd resolution/turn)	2 to each opponent and each of their creatures	yes
Fiery Confluence	3 modes, repeatable: 1 to each creature / 2 to each opponent / destroy artifact	6 to each opponent if all three
Volcanic Vision	MV of the returned card to each creature opponents control	one-sided sweeper
Electrodominance	X to any target + free-cast a spell of MV ≤ X	remember: that free cast forces X=0
Storm King's Thunder	copies your next instant/sorcery X times	force multiplier, not damage
Valakut, the Molten Pinnacle	3 per Mountain entering, needs 5+ other Mountains	any target
Shatterskull Smashing	X split among ≤2 creatures; 2X if X ≥ 6	creatures only
Apex of Power	no damage — +10 mana of one colour and cast 7 free cards	the enabler

All of these double under Solphim except damage to your own permanents.

7. Pump — growing w

Card	Effect on W	Duration
Livaan, Cultist of Tiamat	+MV of each noncreature spell you cast	until end of turn, stacks per spell
Cait Sith, Fortune Teller	+MV of the exiled card, each combat	until end of turn
Blazing Shoal	+X where X = MV of a red card you exile from hand — free	until end of turn
Monstrous Rage	+3/+1, and the Role token is +1/+1 permanently	permanent residue
Blackblade Reforged	+L/+L; equip 3 (legendary), not 7	permanent
Champion's Helm	+2/+2 and hexproof; equip 1	permanent
Commander's Plate	+3/+3 and protection from W/U/B/G; equip 3 for commander	permanent
Forge of Heroes	+1/+1 counter on a commander that entered this turn	permanent
Tyrite Sanctum	2, c: God + a +1/+1 counter; 4, c, sac: indestructible counter	permanent

Livaan is the engine. Cast a mana value 5 spell and Wanda is +5 for the rest of the turn — which is 5 off every subsequent spell, or +1.67 to your X. Cast three spells averaging MV 5 and she's a 17/3 giving 17 off.

Permanent pumps compound across games-long turns. Every Forge/Tyrite/Role counter is a mana saved on every X-spell forever.

8. Copy effects

Card	Copies	Keeps X?	Cost
Pyromancer's Goggles	×1, when its c pays for a red instant/sorcery	✓	free (one per turn)
Increasing Vengeance	×1 from hand, ×2 from the graveyard	✓	c / flashback 3c
Repeated Reverberation	×2	✓	2c
Return the Favor	×1 (and/or redirect)	✓	c + 1 per mode
Storm King's Thunder	×X on your next instant/sorcery	✓	Xc

Copies are not cast. So they do not trigger Thor, Longshot, Fiery Inscription, Livaan, Wiccan, Electro, Birgi or Urabrask. They do trigger Storm-Kiln Artist (which says "cast or copy") and Ashling (same wording).

9. Recursion

Card	What you get back	X-spells?
Past in Flames	flashback on every instant/sorcery in your yard, cost = mana cost	✔ X works
Will of the Jeskai	same, plus a wheel if you control your commander (choose both)	✔ X works
Mizzix's Mastery	exile one I/S from yard, cast a free copy ; overload for all	✗ X = 0
Improvisation Capstone	free-cast off the top; Paradigm = a free copy each first main phase	✗ X = 0
Arcane Bombardment	exiles an I/S at random per turn, copies the whole pile free	✗ X = 0
Underworld Breach (B4)	escape = mana cost + exile 3	✔ X works

10. How to count lethal

Three opponents at 40. You need **120 damage**, or 40 on the one who matters.

Total this turn =	
5X	(Crackle, per target)
+ W	(Chandra's Ignition, each opponent)
+ 2 × S	(Longshot, each opponent)
+ 2 × S	(Fiery Inscription, each opponent)
+ Σ MV	(Thor, focused on one target)
+ 6	(Fiery Confluence, each opponent)
× 2 if Solphim is out	

Three realistic kill lines:

- A. The clean one. Crackle at **X=8** = 40 to each of eight targets. Needs 21–23 mana depending on R. This is the default.
- B. The cheap one. Solphim + Crackle at **X=4** = 40 to each of four targets, for **9 mana** at R=5. Solphim halves the mana requirement of your entire deck.
- C. The Goggles one. Crackle at **X=2** with **Pyromancer's Goggles** = 10 + 10 to the *same* two players = **20 each for ~4 mana**. With Solphim, **40 each — two players dead for four mana**.
- D. No payoff needed. Thor + Longshot + Fiery Inscription out, chain six spells averaging MV 5: $6 \times 5 = 30$ focused (Thor) plus $6 \times 2 + 6 \times 2 = 24$ to *each* opponent. That's 72 spread and 30 focused without casting a single payoff card.

11. Rules gotchas that change the math

- 107.3a — X is the same for every **x** in a cost. Crackle at X=3 is **nine** generic, not three.

While on the stack, MV counts X at its announced value, so Crackle at X=3 is **MV 11** and Wanda discounts it.

- 107.3b — casting "without paying its mana cost" forces **X = 0**. Cost *reduction* is fine.
- 601.2c — the same target can't be chosen twice for one instance of "target." Crackle's targets

are all different. But a **copy is a separate spell**, so it may re-use the same targets.

- 601.2f — total cost = mana cost **or alternative cost**, plus increases, minus reductions.

So Wanda discounts flashback and escape costs too.

- 707.10 — a copy inherits modes, targets, **the value of X**, and alternative costs.
- 500.1 — the postcombat main phase happens **every turn whether or not you attack**. Neheb

triggers regardless.

- Mana empties at the end of each step and phase**. Neheb's pile is postcombat-main only. Electro

and Ashling explicitly override this for **red** mana; Birgi's override lasts until end of turn.

- Ancient Tomb** deals you 2 per activation. At 40 life that's cheap; late, it's real.
- Hexproof and protection do not stop wraths** — a wrath neither targets nor deals damage. Only

Mithril Coat (flash) and **Tyrite Sanctum**'s indestructible counter save Wanda.